

Logistic Regression

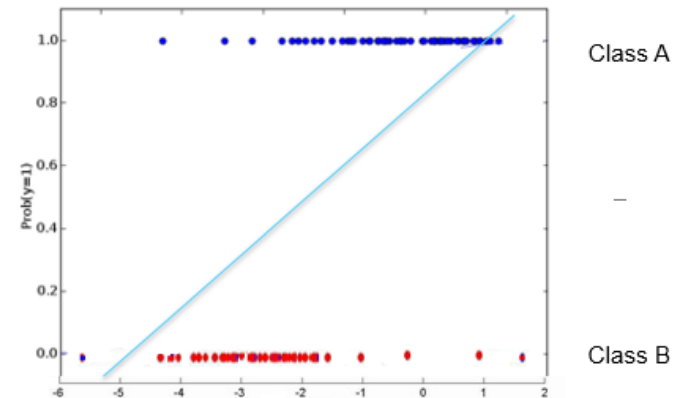
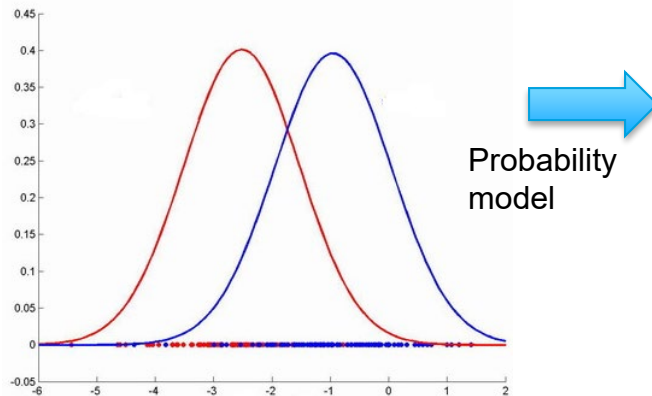
Logistic Regression Model -

- A classification method built on the same concept as linear regression. The response variable is categorical. In its simplest form, the response variable is binary i.e. belongs to one class or the other
- Given the value of predictor (variable x), the model estimates the probability that the new data point belongs to a given class say "A". Probability values can range between 0 and 1.



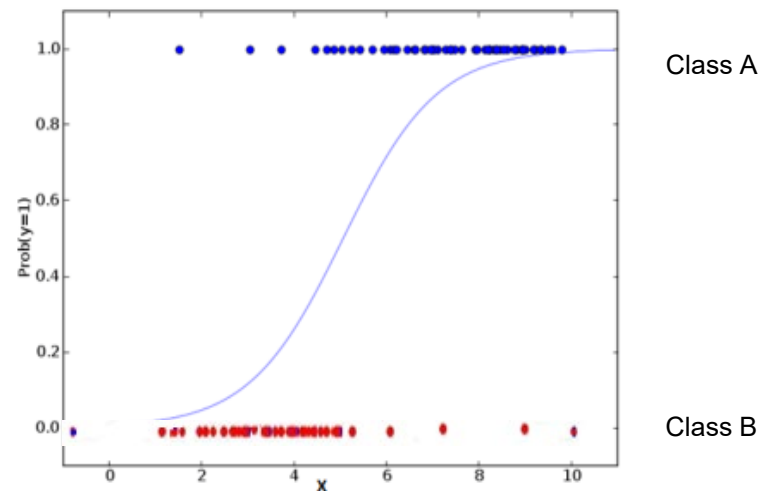
Logistic Regression Model

- c. A new data point (shown with “?”) needs to be classified i.e. does it belong to class A or B.
- d. Given the distribution, closer the point is to the origin, it is unlikely to belong to class A. Farther away it is from the origin, likely it belongs to class A
- e. One can try to fit a simple linear model ($y = mx + c$) where y greater than a threshold means point most probably belongs to class A. The challenge is, for extreme values of x , probability is <0 or >1 which is absurd



Logistic Regression Model -

- f. The linear model is passed to a logistic function $p = 1 / 1 + e^{-t}$ the result of which is values between 0 and 1. Thus p represents probability a data point belongs to class "A" given x



Note: The linear model t (which is of the form $mx + c$), represents logit which is natural log($p / 1-p$) where p is probability that a data point belongs to a class or not.

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Logistic Regression Model -

- g. Uses logloss function to find the best fit line from the infinite possibilities where

$$\log Loss = \frac{-1}{N} \sum_{i=1}^N (y_i (\log p_i) + (1 - y_i) \log(1 - p_i))$$

- h. The objective is to make logLoss as large negative number as possible

- i. There can be four difference cases for the value of y_i and p_i

Case 1: $y_i = 1$, $p_i = \text{High}$, $1 - y_i = 0$, $1 - p_i = \text{Low}$

Correct classification

Case 2: $y_i = 1$, $p_i = \text{Low}$, $1 - y_i = 0$, $1 - p_i = \text{High}$

Incorrect classification

Case 3: $y_i = 0$, $p_i = \text{Low}$, $1 - y_i = 1$, $1 - p_i = \text{High}$

Correct classification

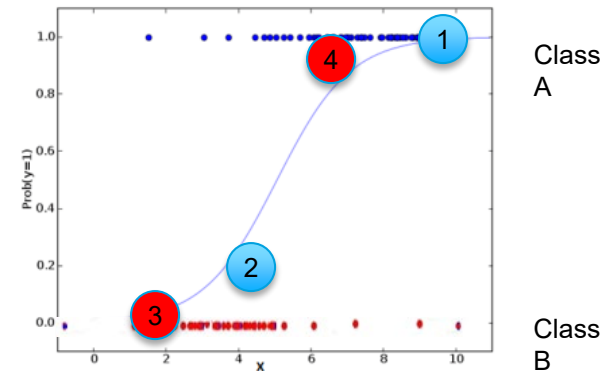
Case 4: $y_i = 0$, $p_i = \text{High}$, $1 - y_i = 1$, $1 - p_i = \text{Low}$

Incorrect classification

- j. Incorrect classification contributes very minimal to the sum while a correct classification contributes large magnitudes

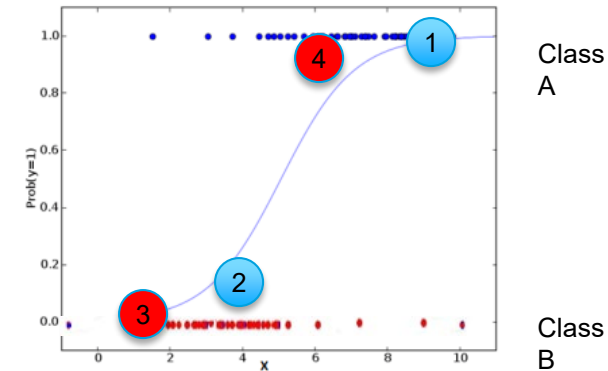
Logistic Regression Model -

- k. In case1 $y = 1$ and $p = \text{high}$ implies that we have got things right!. It significantly inflates the sum because, $Y_i * \log(P_i)$ would be high while the other term in the would be zero since $1 - Y_i = 1 - 1 = 0$.
- l. So more occurrences of Case 1 would inflate the sum and consequently inflate the mean
- m. In case2, $y = 1$ and p is low which is incorrect classification. $p = \text{low}$, $Y_i * \log(P_i)$ would not inflate the sum as much, the second term would be zero since $1 - y_i$ would be zero. So Case 2 would ultimately not affect the sum a lot.
- n. Similarly the occurrences of Case 3 would inflate the sum significantly because first term would be 0 but second term will be high i.e. $(1 - y_i) * \log(1 - p_i)$
- o. Case 4 first term will be 0 while in second term due to high p_i , $(1 - p_i)$ will also be small hence contribution will be small



Logistic Regression Model -

- p. More of Case 1s and Case 3s increase the magnitude of the sum inside the logloss formula and because of the negative sign, make it overall error smaller and smaller
- q. More Case2s and Case4s will not have as bit an impact on the overall value
- r. The objective is to find the logistic curve that makes the overall logloss as negative as possible



Logistic Regression Model -

Advantages -

1. Makes no assumptions about distributions of classes in feature space
2. Easily extended to multiple classes (multinomial regression)
3. Natural probabilistic view of class predictions
4. Quick to train
5. Very fast at classifying unknown records
6. Good accuracy for many simple data sets
7. Resistant to overfitting
8. Can interpret model coefficients as indicators of feature importance

Dis advantages -

1. Constructs linear boundaries

Logistic Regression Model -

Lab- 2- Predict diabetes among Pima Indians

Description – Sample data is available at

<https://archive.ics.uci.edu/ml/machine-learning-databases/pima-indians-diabetes/pima-indians-diabetes.names>

The dataset has 9 attributes listed below

1. Number of times pregnant
2. Plasma glucose concentration a 2 hours in an oral glucose tolerance test
3. Diastolic blood pressure (mm Hg)
4. Triceps skin fold thickness (mm)
5. 2-Hour serum insulin (mu U/ml)
6. Body mass index (weight in kg/(height in m)²)
7. Diabetes pedigree function
8. Age (years)
9. Class variable (0 or 1)

Sol: Logistic Regression - Lima Diabetes.ipynb

Confusion Matrix -

1. A tool to assess the performance of a classification model such as logistic regression model

Total = 231	Predicted 0	Predicted 1	Row Total
Actual 0	134	13	147
Actual 1	38	46	84
Col Total	172	59	231

2. Of the 84 actual diabetes case, the model correctly classified only 46 as diabetic
3. Of the 147 non diabetic cases, the model correctly classified 134 as non-diabetic
4. 13 cases who are normal but identified as diabetic are called Type 1 error
5. 38 cases of diabetic patients identified as normal is Type II error